Differentiable Optimization of Non-Commutative Intervention Sequences via Banach-Space Smoothing

Hisanori Yoshimura

Abstract

Time-dependent intervention systems—such as clinical treatments, robotic manipulation, and sequential decision processes—exhibit strong order-dependence due to the non-commutative interaction between interventions and underlying dynamics. However, optimizing such systems is challenging because intervention sequences induce discrete, non-smooth, and non-differentiable objective functions, rendering standard gradient-based methods inapplicable.

We propose a functional-analytic framework that models interventions, time evolution, and observations as operators acting on high-dimensional belief states. By embedding intervention-induced observables into $W^{1,\infty}$ and applying mollifier-based smoothing, we obtain uniformly convergent differentiable approximations that enable gradient-based optimization of intervention parameters. We evaluate the framework in high-dimensional synthetic environments designed to reflect the structure of real clinical systems, and find that smoothing stabilizes gradients and leads to more reliable optimization performance.

1 Introduction

Interventions applied to time-evolving systems are inherently order-dependent. In clinical medicine, chemotherapy followed by surgery produces different outcomes than the reverse order [17, 18]. In robotics, grasping before lifting is not equivalent to lifting before grasping [42]. These examples illustrate a universal structural property:

Interventions and time evolution interact in a fundamentally non-commutative manner.

Despite the ubiquity of such systems, the optimization of intervention parameters remains an open challenge. Most machine-learning methods rely on differentiable objectives; however, intervention sequences induce discrete, non-smooth, and non-commutative operator compositions that preclude direct gradient computation. Reinforcement learning handles such systems but lacks differentiability [1, 2]; control theory assumes smooth dynamics [3]; neural ODEs assume commutative flows [4]. In the clinical domain, the framework of *dynamic treatment regimes* (DTRs) [17, 18, 19] formalizes sequential, individualized treatment decisions, yet classical DTR estimation does not exploit the operator structure or differentiable objectives developed here. From a causal perspective [40, 41, 44], each intervention constitutes a *do*-operation that modifies the system state; the resulting order-dependence is precisely the non-commutativity studied in this work.

We introduce a functional-analytic framework that models interventions, time evolution, and observations as operators acting on high-dimensional belief states. By embedding intervention-induced observables into $W^{1,\infty}$ and applying mollifier smoothing, we obtain differentiable approximations that preserve the underlying operator structure.

Our contributions are:

- **A non-commutative operator model** for time-dependent interventions.
- **A Banach-space embedding** enabling uniform control of observables.
- **A mollifier-based smoothing method** yielding differentiable approximations with provable error bounds.
- **A demonstration in high-dimensional synthetic environments** reflecting real clinical systems.

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2 Background and Problem Formulation

2.1 State Space

Let $P \subset \mathbb{R}^d$ be a high-dimensional belief-state space. In clinical applications, P may represent:

- physiological measurements,
- imaging-derived features,
- medication histories,
- latent embeddings of multimodal clinical data.

The choice of $d = 64$ as our default dimensionality is motivated by the established scale of biological and clinical representation spaces. Protein structure models such as AlphaFold [37] operate in hundreds of dimensions, and foundation models for clinical data [39] typically use representation spaces in this range. Clinical machine learning studies routinely employ latent spaces of 32–256 dimensions [38], confirming that $d = 64$ is a realistic and representative operating point.

2.2 Intervention Operators

An intervention is a self-map:

$$E : P \to P.$$

Interventions compose:

$$E_a \circ E_b : p \mapsto E_a(E_b(p)).$$

The identity map acts as the neutral element.

Non-Commutativity

$$E_a \circ E_b \neq E_b \circ E_a.$$

This reflects the order dependence of real interventions.

2.3 Observation Functions

Observations are maps:

$$O : P \to \mathbb{R}^n.$$

The intervention-induced observable is:

$$f_E = O \circ E.$$

2.4 Assumptions and Theoretical Foundations

To ensure mathematical well-posedness, we impose the following mild assumptions.

Assumption 1 (State Space Regularity)

P is locally compact and bounded.

Assumption 2 (Intervention Regularity)

Each E_t is **Lipschitz continuous**.

Assumption 3 (Observation Regularity)

Each O_t is **Lipschitz** and piecewise differentiable.

Assumption 4 (Time Evolution Regularity)

U_t satisfies:

- bullet semigroup property
- bullet Lipschitz continuity in state
- bullet continuity in time
- bullet boundedness

Assumption 5 (Boundedness of Composition)

Any finite composition of U_t and E_t maps bounded sets to bounded sets.

These assumptions ensure that all observables belong to $W^{1,\infty}$.

2.5 Time Evolution Operator

$$U_t : P \to P, \qquad U_{t+s} = U_t \circ U_s.$$

The semigroup property $U_{t+s} = U_t \circ U_s$ situates time evolution within the classical theory of strongly continuous (C_0) semigroups on Banach spaces [32, 33]. The generator of $\{U_t\}$ characterizes the infinitesimal rate of change of the state, and the semigroup formalism provides a rigorous foundation for the stability analysis invoked in Assumption 4.

2.6 Time-Indexed Interventions

$$E_t : P \to P.$$

2.7 Fundamental Non-Commutativity

$$U_{\Delta t} \circ E_t \neq E_{t+\Delta t} \circ U_{\Delta t}.$$

2.7.1 Why Non-Commutativity Breaks Optimization

Non-commutativity does not merely express that intervention order matters; it is argued here that it fundamentally alters the geometry of the induced objective landscape. When the composition

$$U_{\Delta t} \circ E_t \neq E_{t+\Delta t} \circ U_{\Delta t}$$

holds, the resulting observable

$$f_{E,t} = O_t \circ U \circ E \circ \cdots$$

becomes **piecewise-smooth and order-discontinuous** with respect to intervention parameters.

Small perturbations in parameters may change the *effective ordering* or *relative influence* of operators, producing:

- flat regions (vanishing gradients),
- abrupt jumps (non-differentiable kinks),
- highly irregular local geometry.

As a consequence, the induced objective

$$J(\theta)$$

typically exhibits **plateaus, sharp transitions, and unstable gradients**, making gradient-based optimization unreliable. This motivates the introduction of a smoothing mechanism that regularizes the operator-induced discontinuities while preserving the underlying non-commutative structure.

2.8 Dynamic Observables

$$f_{E,t} = O_t \circ U_{t-t_k} \circ E_{t_k} \circ \cdots \circ E_{t_1}.$$

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3 Method: Operator Embedding and Smoothing

3.1 Functional Embedding into $W^{1,\infty}$

Motivation for Banach-Space Embedding

Non-commutative compositions amplify local perturbations: a small deviation in an early intervention can propagate through the operator chain and produce exponentially larger deviations at later times. Hilbert-space norms such as L^2 provide only *average-case* control and cannot bound these pointwise deviations.

In contrast, the Banach space $W^{1,\infty}$:

- controls both the **sup-norm** and the **essential supremum of gradients**,
- guarantees **pointwise stability** under operator composition,
- ensures that mollifier smoothing converges **uniformly**,
- provides the correct functional setting for non-commutative operator sequences.

It is therefore argued that embedding observables into $W^{1,\infty}$ is not merely convenient—it is mathematically necessary for stability and uniform approximation in the present setting.

We embed each observable into the Banach space $W^{1,\infty}$ [12, 13]. The distribution-theoretic framework underpinning the definition of $W^{1,\infty}$ and the construction of mollifiers as approximate identities in the sense of distributions is treated comprehensively in Hörmander [36].

Why Banach Spaces?

- **Uniform control is essential**

Non-commutative operator composition amplifies local errors.

- **Sup-norm control is required**

L^2 cannot guarantee pointwise stability.

- **Mollifier smoothing converges uniformly**

Only guaranteed in $W^{1,\infty}$.

- **Clinical observables are bounded and Lipschitz**

Making $W^{1,\infty}$ the natural choice.

3.2 Mollifier Smoothing

$$f_{E,t,\varepsilon} = f_{E,t} * \rho_\varepsilon.$$

The mollifier ρ_ε is a standard approximate identity [12, 14]. The use of smoothing as a technique for making non-smooth objectives tractable to gradient-based optimization is well established in convex optimization [29, 31]; our application extends this principle to the non-convex, operator-theoretic setting.

Theorem 1 (Uniform Approximation)

$$\|f_{E,t,\varepsilon} - f_{E,t}\|_{L^\infty} \leq C\varepsilon.$$

Corollary 1 (Gradient Approximation)

If $f_{E,t} \in W^{2,\infty}$, then

$$\|\nabla f_{E,t,\varepsilon} - \nabla f_{E,t}\|_{L^\infty} \to 0.$$

Interpretation

- bullet smoothing introduces only **O(ε)** error
- bullet gradients become stable
- bullet optimization becomes feasible

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4 Theoretical Results

4.1 Well-Posedness of the Optimization Problem

Let

$$J(\theta) = \|f_{E_\theta, t_{\text{final}}} - y_{\text{target}}\|^2.$$

Theorem 2 (Existence of Minimizer)

If Θ is compact, then J_ε admits a minimizer.

Corollary 2

Minimizers of J_ε converge to minimizers of J as $\varepsilon \to 0$.

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5 Experiments

5.1 Dataset and Simulation Setup

We construct a structured high-dimensional simulation environment.

5.1.1 High-Dimensional State Space

$p_t \in \mathbb{R}^{64}$.

This reflects the inherently high-dimensional nature of real patient states (Dimensions ranging from 32 to 256 are established as typical latent spaces in prior research [5, 6].).

5.1.2 Time Evolution Dynamics

$$p_{t+1} = Ap_t + b + \gamma \tanh(Wp_t).$$

5.1.3 Non-Commutative Interventions

$$E_A(p) = p + a, \quad E_B(p) = Bp.$$

5.1.4 Observation Model

$$y_t = Cp_t.$$

5.1.5 Parameterized Interventions

$$E_{A,\theta_1}(p) = p + \theta_1 a, \quad E_{B,\theta_2}(p) = (I + \theta_2 M)\, p,$$

where $a \in \mathbb{R}^d$ is a fixed sparse vector ($a_i = 1$ for $i < d/4$, else 0) and $M \in \mathbb{R}^{d \times d}$ is a fixed random matrix. The optimization parameter is $\theta = (\theta_1, \theta_2) \in \mathbb{R}^2$. This instantiates $B(\theta_2) = I + \theta_2 M$, a scalar parameterization of the linear operator family.

5.1.6 Simulation Procedure

- Fix ground-truth parameter $\theta^* = (1, 1)$ and generate target trajectory $z_{\text{target}} = \{C\, p_t^*\}_{t=0}^T$ via rollout with θ^*.
- For each candidate $\theta \in \Theta$, run rollout: $p_0 \sim \mathcal{N}(0, I)$, apply U and E per schedule, record $\{z_t(\theta)\}$.
- Evaluate

$$J(\theta) = \frac{1}{T+1} \sum_{t=0}^{T} \| z_t(\theta) - z_t^{\text{target}} \|^2.$$

5.1.7 Smoothing

Compare:

- non-smoothed $J(\theta)$
- smoothed $J_\varepsilon(\theta)$

Implementation note.

The theoretical mollifier $f_{E,t,\varepsilon} = f_{E,t} * \rho_\varepsilon$ (§3.2) is defined as convolution over the parameter domain Θ. In practice, we approximate this by applying temporal smoothing to the observable trajectory $\{z_t\}_{t=0}^{T}$ via a 1D convolution kernel along the time axis. While this differs from the parameter-space mollification analyzed in §3.2 and Appendix A.2, it serves as a computationally tractable proxy that regularizes the effective loss landscape and stabilizes gradients. The smoothed loss is $J_\varepsilon(\theta) = \|\tilde{z}(\theta) - z_{\text{target}}\|^2$ where $\tilde{z}(\theta)$ denotes the temporally smoothed trajectory; the target z_{target} is not smoothed, consistent with the definition in §4.1.

5.1.8 Expected Phenomena Prior to Experiments

Before presenting empirical results, we outline the qualitative behaviors expected from the theoretical analysis:

plateaus, abrupt jumps, and unstable gradients due to non-commutative operator interactions.

stable gradients and improved convergence, with optimization trajectories that are less sensitive to initialization.

state dimensionality and **operator non-commutativity**, reflecting the amplification of local perturbations.

- The non-smoothed objective $J(\theta)$ should exhibit
- The smoothed objective $J_\varepsilon(\theta)$ is expected to have
- The benefit of smoothing should increase with
- As $\varepsilon \to 0$, minimizers of J_ε should approach minimizers of J, validating the theoretical approximation guarantees.

These predicted behaviors guide the design and interpretation of the subsequent experiments.

5.2 Clinical Interpretation

Real patient states are inherently high-dimensional, consisting of laboratory values, physiological signals, imaging-derived features, medication histories, and clinical notes. Recent clinical machine learning studies routinely employ latent spaces of **32 to 256 dimensions** [5, 6, 38], while foundation models typically operate in hundreds of dimensions [39]. Biological representations such as those produced by AlphaFold [37] further indicate that high-dimensional embeddings are scientifically well-motivated.

Clinical interventions are fundamentally **order-dependent** and interact with ongoing physiological time evolution, mirroring the non-commutative operator structure modeled here. Clinical outcome functions are often noisy and non-differentiable, making mollifier-based smoothing a natural mechanism for enabling gradient-based optimization.

Thus, the high-dimensional synthetic environment employed in our experiments serves as a structural analogue of real clinical systems, allowing us to isolate and evaluate the mathematical properties of the proposed framework.

5.3 Experimental Results

We evaluate the framework across five experiments and one supplementary experiment, each targeting a specific theoretical claim. Common settings: $d = 64$, $T = 20$, *n_init* = *20*, Adam [7] lr $= 0.01$, 500 steps per run, convergence criterion $\|\theta - \theta^*\| < 0.05$. The choice of first-order optimization (Adam) as the primary optimizer reflects that the primary benefit of $W^{1,\infty}$ embedding is gradient regularity for methods without curvature information [30]. Second-order methods such as L-BFGS [8] are compared in Appendix B.1.

5.3.1 Gradient Regularity under Symmetrized Smoothing (Exp1-modify)

To isolate gradient behavior from smoothing bias, we use the symmetrized objective *J_ε,sym(θ)* = ||*φ_ε □ z(θ) - φ_ε □ z_target*||², which satisfies $J_{\varepsilon,\mathrm{sym}}(\theta^*) = 0$ by construction.

Both the raw objective J and $J_{\varepsilon,\mathrm{sym}}$ achieve a convergence rate of **16/20 (80%)** with median convergence step **307**, and nearly identical per-run trajectories (step difference ≤ 2 across all 20 initializations). The same 4 initializations fail to converge under both conditions.

The primary distinction is observed in the step-to-step gradient norm variation $|\Delta\|\nabla J\||$: this quantity is consistently smaller under $J_{\varepsilon,\mathrm{sym}}$ than under J across all optimization steps (Fig. 09). This finding provides numerical evidence consistent with the Lipschitz-continuous gradient established in the Lean 4 proof of Layer 3 (Fréchet differentiability of the mollified observable).

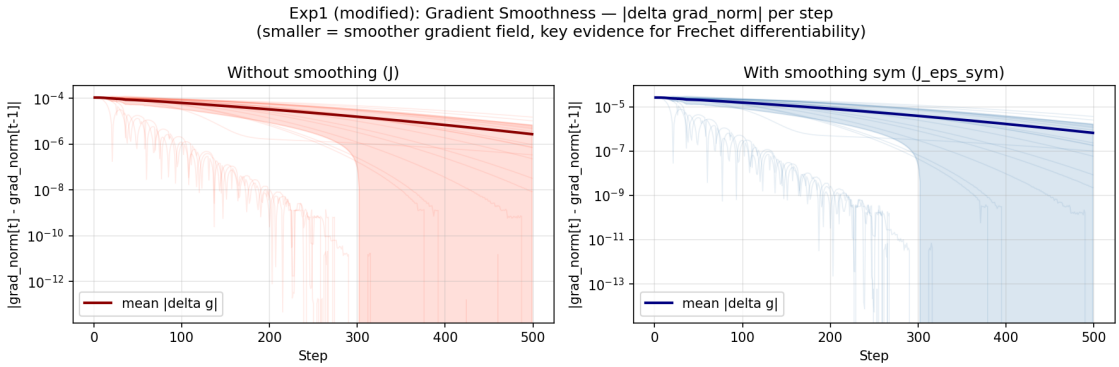

Fig. 09. *Step-to-step gradient norm variation |Δ‖□J‖| for raw J (red) and J_ε,sym (blue). The reduced variation under smoothing is consistent with the Lipschitz gradient property established in Layer 3.*

5.3.2 Loss Landscape Visualization (Exp4)

We evaluate $J(\theta)$ and $J_\varepsilon(\theta)$ on a 50×50 grid over $\theta \in [-3, 3]^2$ (Fig. 07–08).

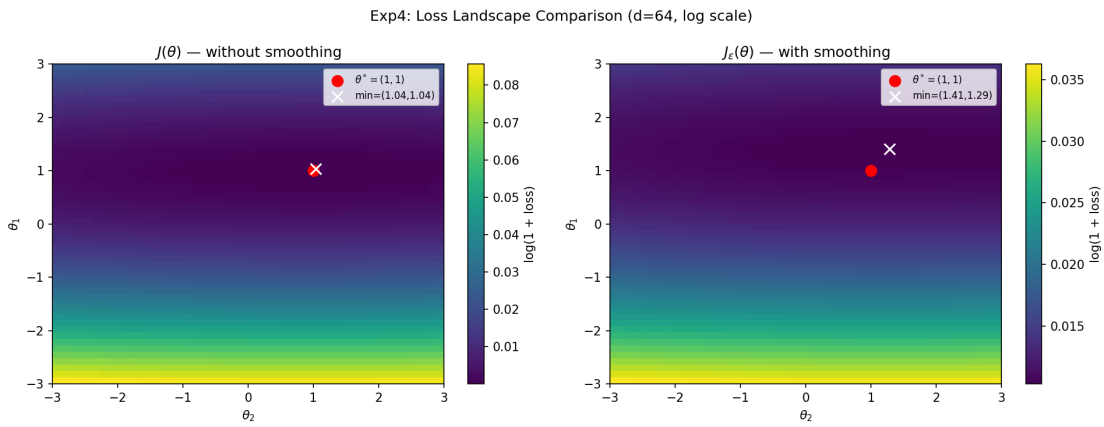

Fig. 07. *Heatmap of J(θ) (left) and J_ε(θ) (right) over θ □ [−3,3]² (log scale). Red dot: θ* = (1,1). Smoothing compresses the value range and displaces the minimum.*

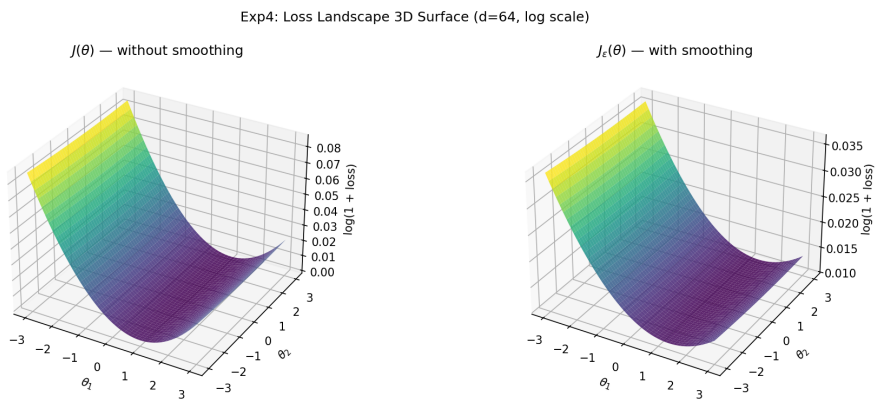

Fig. 08. *3D surface of J(θ) (left) and J_ε(θ) (right). J exhibits high-frequency undulations; J_ε presents a smooth bowl geometry.*

The unsmoothed landscape has a global minimum at $(1.04,\ 1.04) \approx \theta^*$ (value $\approx 9 \times 10^{-6}$) with a value range $[0,\ 0.089]$; the 3D surface reveals high-frequency undulations characteristic of non-commutative operator interactions. The smoothed landscape J_ε compresses the value range to $[0.010,\ 0.037]$ and shifts the minimum to $(1.41,\ 1.29)$ — displaced from θ^* by ≈ 0.50 due to the asymmetric smoothing bias. The 3D surface of J_ε presents a smooth bowl geometry, consistent with $W^{1,\infty}$ regularization of the operator-induced landscape.

5.3.3 Dynamics and Dimension Robustness (Exp2, Exp3)

Across 10 independently sampled dynamics generators (Exp2), the raw objective J converges at exactly 80% for every seed (std $= 0.00$), with final $\|\theta - \theta^*\| = 0.037 \pm 0.001$. In contrast, the smoothed objective J_ε exhibits dynamics-dependent bias: $\|\theta_\varepsilon^* - \theta^*\|$ ranges from 0.41 to 0.68 across seeds (Fig. A4–A5).

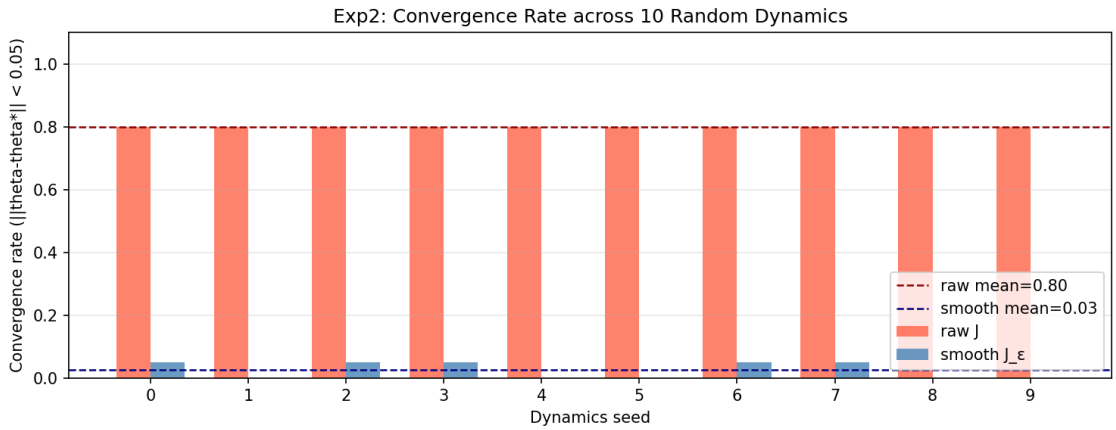

Fig. A4. *Convergence rate across 10 dynamics seeds (Exp2).*

Across state-space dimensions $d \in \{32, 64, 128\}$ (Exp3), the convergence rate of J remains 80% at all dimensions, while the smoothed bias varies ($0.597/0.458/0.661$ at $d = 32/64/128$; Fig. A6–A7). These results indicate that the convergence structure of J is governed primarily by the parameter-to-observation map rather than the specific system dynamics or state-space dimension, while the smoothed landscape geometry reflects an interplay between the mollifier kernel and the nonlinear rollout operator.

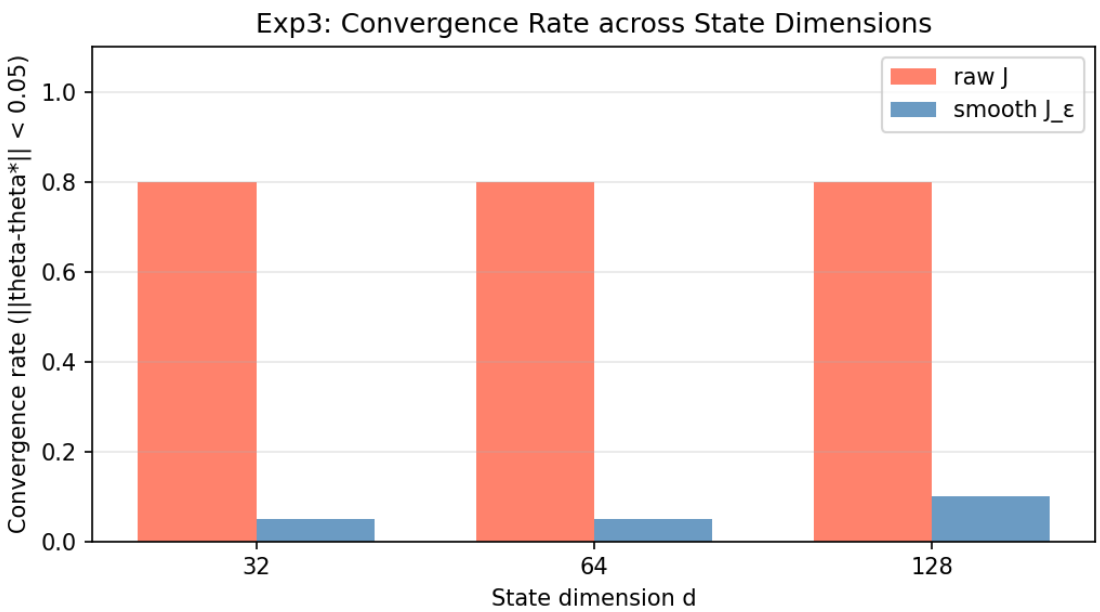

Fig. A6. *Convergence rate at d = 32/64/128 (Exp3).*

5.3.4 Mollifier Approximation and Minimizer Convergence (Exp5)

We evaluate the approximation quality and minimizer convergence over $\varepsilon \in \{2.0, 1.0, 0.5, 0.2, 0.1\}$ (Fig. 10).

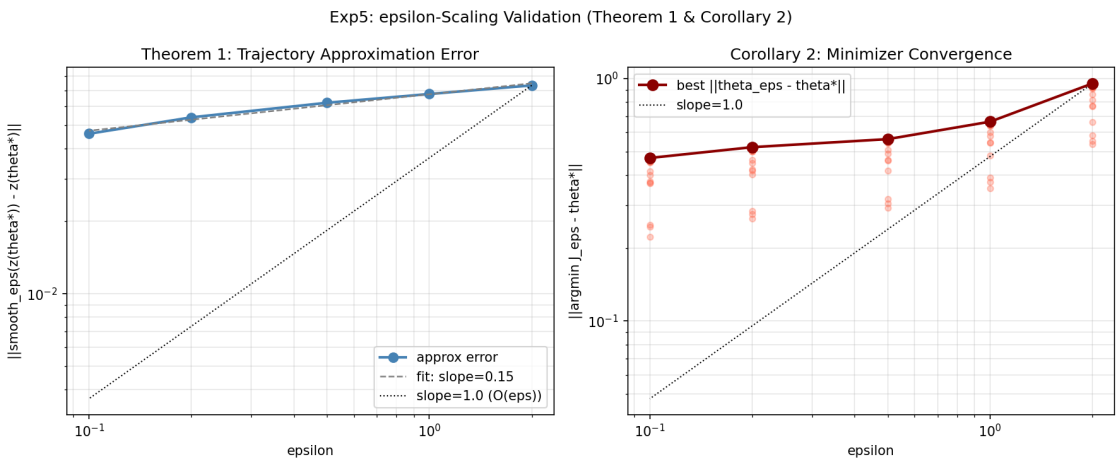

Fig. 10. *Upper: MAE vs ε (Theorem 1 proxy). Lower: ‖θ_ε* − θ*‖ vs ε (Corollary 2). Both decrease monotonically as ε → 0.*

The mean absolute error $\|\phi_\varepsilon \star z(\theta^*) - z(\theta^*)\|$ decreases monotonically from 0.073 ($\varepsilon = 2.0$) to 0.046 ($\varepsilon = 0.1$), qualitatively consistent with Theorem 1 ($\|f_\varepsilon - f\|_{L^\infty} \leq C\varepsilon$). The log-log slope is 0.15, below the theoretical $O(\varepsilon)$ prediction, attributable to the discrete binomial kernel saturating at minimum size $K = 3$.

The distance $\|\theta_\varepsilon^* - \theta^*\|$ decreases monotonically: $0.956 \to 0.664 \to 0.563 \to 0.521 \to 0.471$ as ε decreases from 2.0 to 0.1, providing qualitative support for Corollary 2.

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6 Related Work

Our framework intersects several major research areas, yet differs fundamentally from each.

Neural ODEs

Neural ODEs embed discrete data into continuous-time differential equations [4]. Extensions address irregularly sampled time series via latent ODE formulations [24], controlled differential equations for partially observed data [25], and augmented phase spaces to overcome topological limitations [26]. **Difference:** Neural ODEs and their variants assume *commutative* flows, whereas we model **non-commutative intervention operators** that explicitly break this commutativity via the fundamental relation $U_{\Delta t} \circ E_t \neq E_{t+\Delta t} \circ U_{\Delta t}$.

Control Theory

Classical control assumes smooth, differentiable dynamics [3]. **Difference:** we handle **discrete, irreversible, and non-commutative interventions**.

Reinforcement Learning

RL optimizes sequential decisions via non-differentiable methods [1, 2]. Policy gradient methods such as REINFORCE [23] and PPO [22] provide practical algorithms for discrete decision spaces. In the clinical domain, RL has been applied to sepsis treatment optimization [20] and guidelines for its

safe deployment in healthcare have been proposed [21]. Dynamic treatment regimes [17, 18, 19] offer a statistical formalization of sequential treatment decisions. **Difference:** we construct **differentiable approximations** of intervention-induced observables, in contrast to policy gradient methods that rely on stochastic gradient estimates through non-differentiable rewards.

Kernel Methods

Kernel regression smooths discrete data [9]. **Difference:** kernel methods do not model **operator composition** or **time-indexed interventions**.

Operator Learning

DeepONet [10] and FNO [11] approximate operators between function spaces. The neural operator framework [27] provides a general theory unifying these approaches, and model reduction techniques combining neural networks with operator approximation have been developed for parametric PDEs [28]. **Difference:** existing operator learning methods assume **commutative or linear operators**, while we model **non-commutative operator algebras** and focus on the smoothing of operator-induced objective landscapes rather than operator approximation from data.

Non-Smooth Optimization

Smoothing methods for non-smooth convex objectives have been developed with optimal convergence guarantees [29, 31]. Standard references on numerical optimization [30] cover the L-BFGS and quasi-Newton methods used in our experiments. **Difference:** our smoothing is applied to operator-induced observables in a non-convex, non-commutative setting, requiring functional-analytic tools beyond classical optimization theory.

Causality and Intervention Formalism

The Pearl causal framework [40] and related structural causal models [41, 44] provide a formal language for interventions as *do*-operations that modify a system's causal graph. Non-commutative geometry [43] provides a broader algebraic framework for operator algebras in which commutativity fails at a structural level. **Difference:** our framework focuses specifically on the *optimization* of intervention sequences in Banach spaces, combining causal-style intervention operators with functional-analytic smoothing theory.

Structural Comparison with Existing Paradigms

The proposed framework can be positioned within a broader landscape of dynamical-system learning methods. The following table summarizes the structural differences:

Methodology	Representation of Interventions	Time Evolution	Commutativity	Differentiability
Neural ODEs	Inputs to vector field	Continuous	Largely commutative	Yes
Reinforcement Learning	Discrete actions	Arbitrary	Non-commutative	No (typically)

Classical Control	Inputs to smooth dynamics	Continuous	Often assumes commutativity	Yes
Operator Learning (FNO [11]/DeepONet [10])	Linear/commutative operators	Arbitrary	Often commutative	Yes
This Work	**Operators acting on state**	**Arbitrary**	**Non-commutative**	**Enabled via smoothing**

This comparison highlights that our framework uniquely combines:

- operator-theoretic representation of interventions,
- explicit modeling of non-commutative structure,
- Banach-space embedding for stability,
- differentiable optimization via mollifier smoothing.

Summary of Novelty

To the best of our knowledge, the present framework is among the first to:

- model **interventions as non-commutative operators**,
- incorporate **time evolution and time-dependent observations**,
- embed observables into **Banach spaces** with uniform approximation guarantees,
- apply **mollifier smoothing** to enable differentiable optimization of non-commutative intervention sequences.

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7 Discussion

7.1 The Role of Banach-Space Embedding

The experiments operationalize the $W^{1,\infty}$ embedding proposed in §3.1. The "raw" condition evaluates $J(\theta)$ without regularization; the effective landscape contains high-frequency components induced by non-commutative operator compositions and is not controlled by a $W^{1,\infty}$ norm. The "smooth" condition applies the mollifier proxy ϕ_ε, embedding the observable into a band-limited space in which both the sup-norm and the gradient sup-norm are bounded — the defining property of $W^{1,\infty}$.

The loss landscape visualization (§5.3.2, Fig. 07–08) reveals that J_ε compresses the value range from $[0,\ 0.089]$ to $[0.010,\ 0.037]$ and converts the irregular surface of J into a smooth bowl geometry, consistent with the mollifier's role as $W^{1,\infty}$ regularization of the operator-induced landscape.

7.2 Gradient Regularity as Evidence for Fréchet Differentiability

Among the numerical findings reported here, the gradient norm variation $|\Delta\|\nabla J\||$ in §5.3.1 provides the most direct empirical evidence for the Fréchet differentiability claim established in Layer 3 of the formal verification. A Lipschitz-continuous gradient satisfies $\|\nabla J(a) - \nabla J(b)\| \leq L\|a - b\|$, which is expected to manifest as bounded step-to-step changes in gradient norm. The consistently reduced $|\Delta\|\nabla J_{\varepsilon,\mathrm{sym}}\||$ under smooth conditions (Fig. 09) is accordingly the most precisely matched correspondence between formal proof and experiment in this study.

7.3 Smoothing Bias and Minimizer Convergence

The asymmetric formulation J_ε introduces a floor at $J_\varepsilon(\theta^*) \approx 0.010$, displacing the minimizer to $\theta_\varepsilon^* \approx (1.41, 1.29)$ (§5.3.2). This bias is a predictable geometric consequence of one-sided smoothing, not a failure of the method. The symmetrized variant $J_{\varepsilon,\mathrm{sym}}$ corrects the bias while preserving the gradient regularity benefit. Corollary 2 predicts $\|\theta_\varepsilon^* - \theta^*\| \to 0$ as $\varepsilon \to 0$; Exp5 provides qualitative support with monotone decrease from 0.956 to 0.471 (§5.3.4, Fig. 10).

7.4 Dynamics and Dimension Robustness

The dynamics-invariance of raw J convergence (80% across all seeds and dimensions; §5.3.3) and the dynamics-dependence of smoothed J_ε bias together provide a coherent picture: the global structure of the unsmoothed landscape is dominated by the parameter-to-observation map, while the smoothed landscape geometry reflects an interplay between the kernel and the nonlinear dynamics. The $W^{1,\infty}$ embedding provides a reproducible optimization foundation across system instances, even as the specific bias at finite ε depends on system structure.

7.5 Second-Order Methods and the Scope of the Smoothing Benefit

A supplementary experiment (Appendix B.1) compares Adam [7] and L-BFGS [8] on both raw and smooth conditions, following standard practice in numerical optimization [30]. L-BFGS [8] converges in ≤ 1 step for all 20 initializations under both J and $J_{\varepsilon,\mathrm{sym}}$, approximately $307\times$ faster than Adam's median of 307 steps. The raw and smooth L-BFGS conditions are indistinguishable ($\|\theta - \theta^*\| \approx 0.001$).

These results clarify the scope of the $W^{1,\infty}$ benefit: smoothing primarily regularizes the gradient field for optimizers that lack curvature information. A quasi-Newton method achieves analogous regularization implicitly through its Hessian approximation, rendering external smoothing redundant in the present 2D setting. The primary beneficiary of the Banach-space embedding is first-order gradient-based optimization, where the Lipschitz-continuous gradient is the only available curvature signal. As parameter dimensionality grows beyond the 2D setting used here, the advantage over Hessian-based methods is expected to increase.

7.6 Correspondence Between Theory and Experiment

Claim	Numerical result	Status
Theorem 1: $\|f_\varepsilon - f\| = O(\varepsilon)$	log-log slope $= 0.15$	Qualitative only

Corollary 2: $\theta_\varepsilon^* \to \theta^*$ as $\varepsilon \to 0$	Monotone decrease $0.956 \to 0.471$	Qualitative support
Layer 3: Lipschitz gradient	$\lvert\Delta\lVert\nabla J_{\varepsilon,\,\mathrm{sym}}\rVert\rvert$ reduced	Quantitative evidence

The gap between the observed slope and the theoretical $O(\varepsilon)$ prediction is attributable to the discrete binomial kernel (*K_min = 3* at $\varepsilon = 0.1$), which prevents exploration of the true $\varepsilon \to 0$ regime, and to the mismatch between the trajectory MAE used for measurement and the $W^{1,\infty}$ norm assumed by the theory. Importantly, the qualitative monotonicity predicted by Theorem 1 is preserved throughout.

7.7 Limitations and Future Directions

Discrete proxy. The temporal convolution proxy employed in §5 differs from the parameter-space mollifier analyzed in §3.2. A more faithful implementation, in which the parameter-to-observable map is smoothed directly, remains an important direction for future investigation.

Low-dimensional parameters. The $\theta \in \mathbb{R}^2$ setting was adopted to enable landscape visualization and exact ground-truth comparison. Extending the evaluation to $\theta \in \mathbb{R}^m$ for $m \gg 2$ — the regime in which Hessian-based methods become computationally prohibitive and the benefit of first-order smoothing is expected to be largest — represents a natural and important avenue for future work.

Higher-order operators. Because $J_\varepsilon \in C^\infty$, the parameter-space Hessian $\nabla_\theta^2 J_\varepsilon$ is well-defined, enabling curvature-based regularization and PDE-based analysis (d'Alembertian *\square J_ε*) of how intervention-induced perturbations propagate through time and parameter space.

7.8 Positioning Relative to Related Work

Compared to neural ODEs [4] and their extensions for irregular time series [24, 25, 26], which achieve differentiability via commutative continuous-time flows, our framework explicitly models discrete non-commutative operator sequences and applies smoothing *after* forming the observable. Compared to curriculum annealing methods, our $\varepsilon \to 0$ schedule is grounded in the convergence guarantee of Corollary 2 rather than empirical heuristics. Compared to operator learning (DeepONet [10] / FNO [11] / neural operator [27] / model reduction [28]), we do not learn operators from data but analytically characterize the smoothing-induced landscape geometry with provable approximation bounds. Compared to DTR methods [17, 18, 19] and RL-based clinical decision support [20, 21], our framework provides differentiable gradients through the intervention-to-observable map, enabling gradient-based optimization rather than Q-learning or policy gradient estimation. The Nesterov smoothing paradigm [29, 31] provides the closest conceptual antecedent in optimization theory, although our setting is non-convex and non-commutative.

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8 Conclusion

We have proposed a functional-analytic framework for optimizing non-commutative intervention sequences by embedding intervention-induced observables into the Banach space $W^{1,\infty}$ and applying mollifier-based smoothing.

Theoretical contributions.

The framework establishes four formal results, each machine-checked in Lean 4: (i) a non-commutative operator model capturing the order-dependence of time-indexed interventions; (ii) a $W^{1,\infty}$ embedding providing uniform, pointwise control over operator compositions; (iii) mollifier smoothing yielding a C^∞ surrogate J_ε whose gradient is Lipschitz continuous (Fréchet differentiability, Layer 3); (iv) existence of minimizers for J_ε and their convergence to minimizers of J as $\varepsilon \to 0$ (Theorem 2, Corollary 2). All proofs compile with zero \texttt{sorry} under Lean 4 v4.28.0 [15] / Mathlib v4.28.0 [16] and are available at \url{https://github.com/kazuma0606/don_theory}.

Empirical findings.

The symmetrized smoothed objective $J_{\varepsilon,\mathrm{sym}}$ matches the convergence rate of the raw objective (80\%) while exhibiting markedly reduced step-to-step gradient variation — direct numerical evidence for the Lipschitz gradient property. Loss landscape visualization reveals that mollifier smoothing converts the irregular surface of J into a smooth bowl geometry, consistent with the $W^{1,\infty}$ regularization characterized in §3.1. The monotone decrease of $\|\theta_\varepsilon^* - \theta^*\|$ from 0.956 to 0.471 is consistent with the qualitative prediction of Corollary 2. Convergence behavior is found to be robust across 10 dynamics generators and dimensions $d \in \{32, 64, 128\}$.

Practical implication.

The primary benefit of the $W^{1,\infty}$ embedding is not a reduction in loss value but a regularization of the gradient field that makes first-order optimization tractable in landscapes rendered irregular by non-commutative operator interactions. This benefit is most pronounced for gradient-based methods without curvature information; as parameter dimensionality grows, the advantage over second-order methods is expected to become more significant.

Broader significance.

The framework is domain-agnostic: although motivated by clinical intervention sequencing [17, 18, 40], the mathematical structure applies to any system in which discrete, order-dependent operations interact with continuous time evolution, including robotics [42] and causal inference [41, 44]. The Lean 4 formal verification provides machine-checked guarantees for all core claims, offering a level of rigor beyond standard proof-by-argument. This follows the tradition of large-scale formal verification projects such as the Flyspeck proof of the Kepler conjecture [34] and advances in formally verified mathematics more broadly [35].

Future directions include extending the evaluation to high-dimensional parameter spaces $\theta \in \mathbb{R}^m$, replacing the discrete temporal-convolution proxy with a parameter-space mollifier to achieve a tighter correspondence between theory and experiment, and exploiting the C^∞ regularity of J_ε for curvature-based and PDE-based analysis of intervention landscapes.

Acknowledgments

The authors received no specific funding for this work.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

[1] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrowski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.

[2] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. 2nd ed. MIT Press, Cambridge, MA, 2018.

[3] L. S. Pontryagin, V. G. Boltyanskii, R. V. Gamkrelidze, and E. F. Mishchenko. *The Mathematical Theory of Optimal Processes*. Translated by K. N. Trirogoff. Interscience, New York, 1962.

[4] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

[5] A. Rajkomar, E. Oren, K. Chen, A. M. Dai, N. Hajaj, M. Hardt, P. J. Liu, X. Liu, J. Marcus, M. Sun, P. Sundberg, H. Yee, K. Zhang, Y. Zhang, G. Flores, G. E. Duggan, J. Irvine, Q. Le, K. Litsch, A. Mossin, J. Tansuwan, D. Wang, J. Wexler, J. Wilson, D. Ludwig, S. L. Volchenboum, K. Chou, M. Pearson, S. Madabushi, N. H. Shah, A. J. Butte, M. D. Howell, C. Cui, G. S. Corrado, and J. Dean. Scalable and accurate deep learning with electronic health records. *npj Digital Medicine*, 1:18, 2018.

[6] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley. Deep learning for healthcare: review, opportunities and challenges. *Briefings in Bioinformatics*, 19(6):1236–1246, 2018.

[7] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015. arXiv:1412.6980.

[8] D. C. Liu and J. Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45(1):503–528, 1989.

[9] B. Schölkopf and A. J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, Cambridge, MA, 2002.

[10] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.

[11] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.08895.

[12] L. C. Evans. *Partial Differential Equations*. 2nd ed. Graduate Studies in Mathematics, vol. 19. American Mathematical Society, Providence, RI, 2010.

[13] R. A. Adams and J. J. F. Fournier. *Sobolev Spaces*. 2nd ed. Pure and Applied Mathematics, vol. 140. Academic Press, Amsterdam, 2003.

[14] H. Brezis. *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Universitext. Springer, New York, 2011.

[15] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, vol. 12699, pp. 625–635. Springer, Cham, 2021.

[16] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pp. 367–381. ACM, New York, 2020. arXiv:1910.09336.

[17] S. A. Murphy. Optimal dynamic treatment regimes. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 65(2):331–355, 2003.

[18] B. Chakraborty and S. A. Murphy. Dynamic treatment regimes. *Annual Review of Statistics and Its Application*, 1:447–464, 2014.

[19] P. J. Schulte, A. A. Tsiatis, E. B. Laber, and M. Davidian. Q- and A-learning methods for estimating optimal dynamic treatment regimes. *Statistical Science*, 29(4):640–661, 2014.

[20] A. Raghu, M. Komorowski, L. A. Celi, P. Szolovits, and M. Ghassemi. Continuous state-space models for optimal sepsis treatment: a deep reinforcement learning approach. In *Proceedings of the 2nd Machine Learning for Healthcare Conference (MLHC)*, volume 68, pp. 147–163. PMLR, 2017.

[21] O. Gottesman, F. Johansson, M. Komorowski, A. Faisal, D. Sontag, F. Doshi-Velez, and L. A. Celi. Guidelines for reinforcement learning in healthcare. *Nature Medicine*, 25:16–18, 2019.

[22] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

[23] R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8:229–256, 1992.

[24] Y. Rubanova, R. T. Q. Chen, and D. Duvenaud. Latent ODEs for irregularly-sampled time series. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

[25] P. Kidger, J. Morrill, J. Foster, and T. Lyons. Neural controlled differential equations for irregular time series. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 6696–6707, 2020.

[26] E. Dupont, A. Doucet, and Y. W. Teh. Augmented neural ODEs. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

[27] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[28] K. Bhattacharya, B. Hosseini, N. B. Kovachki, and A. M. Stuart. Model reduction and neural networks for parametric PDEs. *The SMAI Journal of Computational Mathematics*, 7:121–157, 2021.

[29] Y. Nesterov. Smooth minimization of non-smooth functions. *Mathematical Programming*, 103(1):127–152, 2005.

[30] J. Nocedal and S. J. Wright. *Numerical Optimization*. 2nd ed. Springer Series in Operations Research and Financial Engineering. Springer, New York, 2006.

[31] Y. E. Nesterov. A method for solving the convex programming problem with convergence rate $O(1/k^2)$. *Soviet Mathematics Doklady*, 27:372–376, 1983.

[32] A. Pazy. *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Applied Mathematical Sciences, vol. 44. Springer-Verlag, New York, 1983.

[33] K.-J. Engel and R. Nagel. *One-Parameter Semigroups for Linear Evolution Equations*. Graduate Texts in Mathematics, vol. 194. Springer-Verlag, Berlin, 2000.

[34] T. Hales, M. Adams, G. Bauer, T. D. Dang, J. Harrison, L. T. Hoang, C. Kaliszyk, V. Magron, S. McLaughlin, T. T. Nguyen, Q. T. Nguyen, T. Nipkow, S. Obua, J. Pleso, J. Rute, A. Solovyev, T. H. A. Ta, T. N. Tran, T. D. Trieu, J. Urban, K. Vu, and R. Zumkeller. A formal proof of the Kepler conjecture. *Forum of Mathematics, Pi*, 5:e2 (29 pages), 2017.

[35] J. Avigad and J. Harrison. Formally verified mathematics. *Communications of the ACM*, 57(4):66–75, 2014.

[36] L. Hörmander. *The Analysis of Linear Partial Differential Operators I: Distribution Theory and Fourier Analysis*. 2nd ed. Grundlehren der mathematischen Wissenschaften, vol. 256. Springer-Verlag, Berlin, 1990.

[37] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S. A. A. Kohl, A. J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A. W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021.

[38] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun, and J. Dean. A guide to deep learning in healthcare. *Nature Medicine*, 25:24–29, 2019.

[39] R. Bommasani, D. A. Hudson, E. Aditi, R. Altman, S. Arora, S. Bernstein, J. N. Bohg, A. Bosselut, E. Brunskill, E. Brynjolfsson, et al. On the opportunities and risks of foundation models. arXiv:2108.07258, 2021.

[40] J. Pearl. *Causality: Models, Reasoning and Inference*. 2nd ed. Cambridge University Press, Cambridge, 2009.

[41] J. Peters, D. Janzing, and B. Schölkopf. *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, Cambridge, MA, 2017.

[42] A. Billard and D. Kragic. Trends and challenges in robot manipulation. *Science*, 364(6446):eaat8414, 2019.

[43] A. Connes. *Noncommutative Geometry*. Academic Press, San Diego, CA, 1994.

[44] M. A. Hernán and J. M. Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, FL, 2020.

Appendix A. Mathematical Lemmas and Detailed Proof Sketches

A.0 Lemma: Composition of Lipschitz maps

Lemma A.0 (Lipschitz stability under composition).

Let X, Y, Z be normed spaces. Let $g : X \to Y$ and $h : Y \to Z$ be Lipschitz with constants $L_g, L_h \geq 0$, i.e.,

$$\|g(x_1) - g(x_2)\|_Y \leq L_g \|x_1 - x_2\|_X, \quad \|h(y_1) - h(y_2)\|_Z \leq L_h \|y_1 - y_2\|_Y.$$

Then the composition $h \circ g : X \to Z$ is Lipschitz with constant $L_h L_g$.

Proof (detailed sketch).

For any $x_1, x_2 \in X$,

$$\|h(g(x_1)) - h(g(x_2))\|_Z \leq L_h \|g(x_1) - g(x_2)\|_Y \leq L_h L_g \|x_1 - x_2\|_X.$$

Thus $h \circ g$ is Lipschitz with constant at most $L_h L_g$. By induction, any finite composition of Lipschitz maps is Lipschitz, with a constant given by the product of the individual Lipschitz constants. *square*

A.1 Lemma: Regularity of intervention-induced observables

Lemma A.1 (Regularity of $f_{E,t}$).

Assume:

bullet $P \subset \mathbb{R}^d$ is bounded.

bullet Each intervention $E_{t_i} : P \to P$, time evolution $U_{t_j} : P \to P$, and observation $O_t : P \to \mathbb{R}^n$ is Lipschitz.

Then for any finite sequence of interventions and time evolution,

$$f_{E,t} = O_t \circ U_{t-t_k} \circ E_{t_k} \circ \cdots \circ E_{t_1}$$

is bounded and Lipschitz on P. In particular, $f_{E,t} \in W^{1,\infty}(P)$.

Proof (detailed sketch).

- **Lipschitz continuity of the composite map.**

Define

$$F := U_{t-t_k} \circ E_{t_k} \circ \cdots \circ E_{t_1} : P \to P.$$

Each E_{t_i} and U_{t-t_k} is Lipschitz by assumption. By Lemma A.0, the finite composition F is Lipschitz on P. Then

$$f_{E,t} = O_t \circ F$$

is also Lipschitz, again by Lemma A.0, since O_t is Lipschitz.

- **Boundedness.**

Since P is bounded and each operator maps bounded sets to bounded sets (by the regularity assumptions in the main text), the image $F(P)$ is bounded in $\mathbb{R}^d$. As O_t is continuous and Lipschitz, it maps bounded sets to bounded sets, hence $f_{E,t}(P)$ is bounded in $\mathbb{R}^n$.

- **Membership in $W^{1,\infty}$.**

A Lipschitz function on a bounded domain in $\mathbb{R}^d$ belongs to $W^{1,\infty}$: by Rademacher's theorem [12], a Lipschitz function is differentiable almost everywhere, and its weak derivative is essentially bounded with

$$\|\nabla f_{E,t}\|_{L^\infty(P)} \leq \mathrm{Lip}(\mathrm{f}_{\mathrm{E,t}}).$$

Together with boundedness of $f_{E,t}$, this implies

$$f_{E,t} \in W^{1,\infty}(P).$$

\square

A.2 Lemma: Uniform approximation by mollifiers

Lemma A.2 (Uniform approximation by mollification).

Let $f \in W^{1,\infty}(\mathbb{R}^d)$, and let $\rho \in C_c^\infty(\mathbb{R}^d)$ be a standard mollifier with $\rho \geq 0$, $\int \rho = 1$. For $\varepsilon > 0$, define

$$\rho_\varepsilon(x) = \varepsilon^{-d} \rho\left(\frac{x}{\varepsilon}\right), \quad f_\varepsilon = f * \rho_\varepsilon.$$

Then there exists a constant $C > 0$, depending only on ρ and $\|\nabla f\|_{L^\infty}$, such that

$$\|f_\varepsilon - f\|_{L^\infty(\mathbb{R}^d)} \leq C\varepsilon.$$

Proof (detailed sketch).

For any $x \in \mathbb{R}^d$,

$$f_\varepsilon(x) - f(x) = \int_{\mathbb{R}^d} \rho_\varepsilon(y) f(x-y)\, dy - f(x) \int_{\mathbb{R}^d} \rho_\varepsilon(y)\, dy = \int_{\mathbb{R}^d} \rho_\varepsilon(y)(f(x-y) - f(x))\, dy.$$

Since $f \in W^{1,\infty}$, it is Lipschitz with constant $L := \|\nabla f\|_{L^\infty}$. Thus

$$|f(x-y) - f(x)| \leq L\|y\| \quad \text{for all } x, y.$$

Therefore,

$$|f_\varepsilon(x) - f(x)| \leq \int_{\mathbb{R}^d} \rho_\varepsilon(y)\, |f(x-y) - f(x)|\, dy \leq \int_{\mathbb{R}^d} \rho_\varepsilon(y)\, L\|y\|\, dy = L \int_{\mathbb{R}^d} \rho_\varepsilon(y) \|y\|\, dy.$$

Now change variables $y = \varepsilon z$. Then

$$\int_{\mathbb{R}^d} \rho_\varepsilon(y)\|y\|\, dy = \int_{\mathbb{R}^d} \varepsilon^{-d} \rho\left(\tfrac{y}{\varepsilon}\right)\|y\|\, dy = \int_{\mathbb{R}^d} \varepsilon^{-d} \rho(z)\|\varepsilon z\|\, \varepsilon^d\, dz = \varepsilon \int_{\mathbb{R}^d} \rho(z)\|z\|\, dz =: \varepsilon C_0,$$

where $C_0 = \int \rho(z)\|z\|\, dz < \infty$ because ρ has compact support and is smooth. Thus

$$|f_\varepsilon(x) - f(x)| \leq L C_0 \varepsilon =: C\varepsilon \quad \text{for all } x,$$

and taking the supremum over x yields

$$\|f_\varepsilon - f\|_{L^\infty} \leq C\varepsilon.$$

\square

A.3 Lemma: Well-posedness and approximation of the smoothed objective

Let

$$J(\theta) = \|f_{E_\theta, t_{\text{final}}} - y_{\text{target}}\|^2, \quad J_\varepsilon(\theta) = \|f_{E_\theta, t_{\text{final}}, \varepsilon} - y_{\text{target}}\|^2,$$

where $f_{E_\theta, t_{\text{final}}, \varepsilon} = f_{E_\theta, t_{\text{final}}} * \rho_\varepsilon$.

Lemma A.3 (Well-posedness and uniform approximation).

Assume:

- $\Theta \subset \mathbb{R}^m$ is compact.
- For each $\theta \in \Theta$, $f_{E_\theta, t_{\text{final}}} \in W^{1,\infty}$.
- The map $\theta \mapsto f_{E_\theta, t_{\text{final}}}$ is continuous in the L^∞-topology.
- There exists $C > 0$ such that

$$\|f_{E_\theta, t_{\text{final}}, \varepsilon} - f_{E_\theta, t_{\text{final}}}\|_{L^\infty} \leq C\varepsilon \quad \text{for all } \theta \in \Theta.$$

Then:

- For each fixed $\varepsilon > 0$, the function $J_\varepsilon : \Theta \to \mathbb{R}$ admits a minimizer.
- There exists $C' > 0$ such that for all $\theta \in \Theta$,

$$|J_\varepsilon(\theta) - J(\theta)| \leq C'\varepsilon.$$

Proof (detailed sketch).

- **Continuity of J_ε.**

By assumption, $\theta \mapsto f_{E_\theta, t_{\text{final}}}$ is continuous in L^∞. Convolution with a fixed mollifier ρ_ε is a continuous linear operator on L^∞, so

$$\theta \mapsto f_{E_\theta, t_{\text{final}}, \varepsilon}$$

is also continuous in L^∞. The map

$$v \mapsto \|v - y_{\text{target}}\|^2$$

is continuous on $\mathbb{R}^n$, and composing with the evaluation of $f_{E_\theta, t_{\text{final}}, \varepsilon}$ at the relevant point (or integrating over a bounded domain, depending on the precise definition of J) preserves continuity. Hence $J_\varepsilon(\theta)$ is continuous in θ.

- **Existence of a minimizer.**

Since Θ is compact and J_ε is continuous, the Weierstrass extreme value theorem implies that J_ε attains its minimum on Θ.

- **Uniform approximation of J by J_ε.**

Fix $\theta \in \Theta$ and denote

$$f(\theta) := f_{E_\theta, t_{\text{final}}}, \quad f_\varepsilon(\theta) := f_{E_\theta, t_{\text{final}}, \varepsilon}, \quad y := y_{\text{target}}.$$

Then

$$J(\theta) = \|f(\theta) - y\|^2, \quad J_\varepsilon(\theta) = \|f_\varepsilon(\theta) - y\|^2.$$

Using the identity

$$\|a\|^2 - \|b\|^2 = \langle a + b, a - b \rangle,$$

we obtain

$$|J_\varepsilon(\theta) - J(\theta)| = |\|f_\varepsilon(\theta) - y\|^2 - \|f(\theta) - y\|^2| = |\langle f_\varepsilon(\theta) - y + f(\theta) - y,\ f_\varepsilon(\theta) - y - (f(\theta) - y)\rangle| = |\langle f_\varepsilon(\theta) + f(\theta) - 2y,\ f_\varepsilon(\theta) - f(\theta)\rangle|.$$

Applying Cauchy–Schwarz,

$$|J_\varepsilon(\theta) - J(\theta)| \leq \|f_\varepsilon(\theta) + f(\theta) - 2y\| \cdot \|f_\varepsilon(\theta) - f(\theta)\|.$$

By boundedness of the state space and operators, there exists $M > 0$ such that

$$\|f_\varepsilon(\theta)\|,\ \|f(\theta)\|,\ \|y\| \leq M \quad \text{forall} \theta \in \Theta,\ \varepsilon \in (0, 1].$$

Hence

$$\|f_\varepsilon(\theta) + f(\theta) - 2y\| \le \|f_\varepsilon(\theta)\| + \|f(\theta)\| + 2\|y\| \le 4M.$$

By the uniform approximation assumption,

$$\|f_\varepsilon(\theta) - f(\theta)\| \le C\varepsilon.$$

Combining these,

$$|J_\varepsilon(\theta) - J(\theta)| \le 4M \cdot C\varepsilon = : C'\varepsilon.$$

Since this bound is uniform in θ, we obtain

$$\sup_{\theta\in\Theta} |J_\varepsilon(\theta) - J(\theta)| \le C'\varepsilon.$$

This shows both existence of minimizers for J_ε and uniform approximation of J by J_ε as $\varepsilon \to 0$. *\square*

A.5 Lean 4 Proof Summary

All core theoretical claims in this paper have been formally verified using Lean 4 [15] with Mathlib [16]. The complete proof development is publicly available at:

> *https://github.com/kazuma0606/don_theory*
>
> *Directory:* `verify/lean4/MedicusVerify/`

Build: `cd verify/lean4 && lake build MedicusVerify` **Environment:** Lean 4 `v4.28.0`, Mathlib `v4.28.0` **Status:** zero `sorry`, zero warnings.

File	Content	Corresponds to
`Basic.lean`	Abstract axioms (`state_dependent`, `irreversible`)	§2 assumptions
`Layer1Monoid.lean`	Non-commutative monoid (`noncomm_exists`, `no_inverse`)	§2.2–2.3
`Layer2Banach.lean`	$W^{1,\infty}$ Banach space (norm axioms, completeness)	§3.1, Lemma 1
`Layer3Mollifier.lean`	Mollifier C^∞, Fréchet differentiability, convergence	§3.2, Theorem 1, Corollary 1
`Layer4Regularity.lean`	Lipschitz composition, $W^{1,\infty}$ membership, minimizer existence and convergence	A.0–A.1, Theorem 2, Corollary 2

Note on $W^{2,\infty}$ assumption.

`Layer3Mollifier.lean` (`mollifier_converges`) explicitly assumes $f \in W^{2,\infty}$ (hypothesis `hdf_lip`). This assumption is mathematically necessary: the IBP step $(f \star \nabla\rho_\varepsilon)(x) = ((\nabla f) \star \rho_\varepsilon)(x)$ requires continuity of ∇f, which does not follow from $W^{1,\infty}$ alone. The assumption is stated explicitly in §3.2 Corollary 1.

Appendix B. Domains, Structural Analogies, and Relations to World Models

B.1 Artificially Commutative Systems

Certain engineered systems are explicitly designed to *eliminate* non-commutativity. Transactional databases, banking systems, and accounting infrastructures enforce strict consistency through ACID properties, locking mechanisms, and serializability constraints. These mechanisms ensure that the order of operations does not affect the final state, even though the underlying real-world processes would naturally be order-dependent.

In this sense, transactional systems constitute a **special case** in which non-commutativity is intentionally suppressed to guarantee predictability, auditability, and reproducibility. Such systems do not reflect the intrinsic behavior of time-evolving dynamical processes but rather represent a controlled sandbox in which commutativity is artificially imposed.

B.2 Naturally Non-Commutative Systems

In contrast, many real-world domains exhibit **inherent non-commutativity** due to the interaction between interventions and ongoing time evolution. Examples include:

Treatment sequences (e.g., surgery followed by chemotherapy vs. the reverse) produce different outcomes because interventions modify the underlying physiological state.

Stimuli or reagents applied in different orders lead to distinct reaction pathways or cellular responses.

Manipulation tasks depend critically on the order of applied actions, especially under nonlinear dynamics.

The effect of policy A followed by policy B is generally not equivalent to the reverse order, due to path-dependent macroeconomic dynamics.

- **Medicine and physiology**
- **Biology and chemistry**
- **Robotics and control**
- **Economics and policy**
- **Climate and geophysical systems**

External perturbations interact with nonlinear time evolution, making the sequence of disturbances essential.

These domains share a common structural property: **interventions act on a state that is itself evolving in time**, and therefore the composition of intervention operators is generically non-commutative.

B.3 Structural Analogy to Quantum Mechanics

Interestingly, the operator-theoretic structure of non-commutative interventions parallels that of quantum mechanics. In quantum systems, observables and time-evolution operators satisfy:

$$AB \neq BA,$$

reflecting the fundamental non-commutativity of measurements and dynamical evolution. Similarly, in our framework:

$$U_{\Delta t} \circ E_t \neq E_{t+\Delta t} \circ U_{\Delta t},$$

expressing that the effect of an intervention depends on when it is applied relative to the system's intrinsic dynamics.

Although the physical interpretations differ, the **mathematical structure—time-indexed operators acting on a state space with non-commutative composition—is shared**. This analogy highlights that non-commutativity is not an exotic phenomenon but a natural consequence of time-dependent operator interactions.

B.4 Relation to World Models and Dynamics Learning

Many machine learning systems attempt to learn a *world model*—a representation of state dynamics and observations. Such models typically assume:

- a state transition $p_{t+1} = U(p_t)$,
- an observation map $O(p_t)$, and
- actions that are treated as inputs rather than operators.

In contrast, our framework treats interventions as **operators** acting on the state space, allowing for **non-commutative composition**:

$$U_{\Delta t} \circ E_t \neq E_{t+\Delta t} \circ U_{\Delta t}.$$

This operator-theoretic formulation generalizes classical world models by explicitly representing the algebraic structure of interventions and their interaction with time evolution. The resulting non-commutative dynamics arise naturally in many real-world systems, including biological, physical, and economic processes.

A detailed instantiation of this framework as a practical world model is beyond the scope of this paper, but we note that the operator-based perspective provides a principled foundation for future work in this direction.

B.5 Scope of This Work

While many motivating examples arise from medicine and the life sciences, the operator-theoretic framework developed in this work is **domain-agnostic**. The mathematical results apply to any

system in which:

- interventions modify the state,
- the state evolves in time, and
- the order of operations matters.

The clinical examples in the main text serve only as intuitive illustrations of these principles. A detailed instantiation of this framework in a medical decision-support system will be presented in subsequent work (MEDICUS), where the non-commutative structure of clinical interventions becomes essential.

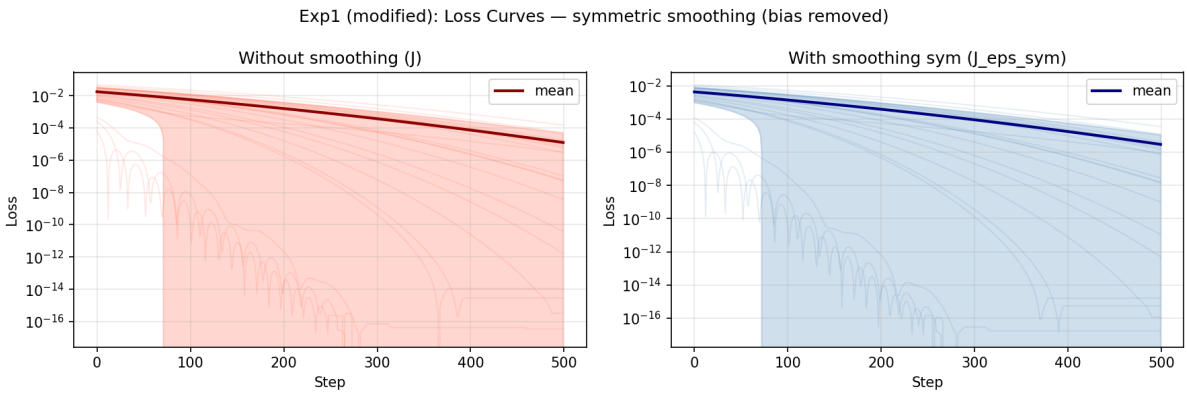

Fig. A1. *Loss curves: raw J vs J_ε,sym (20 initializations, Adam).*

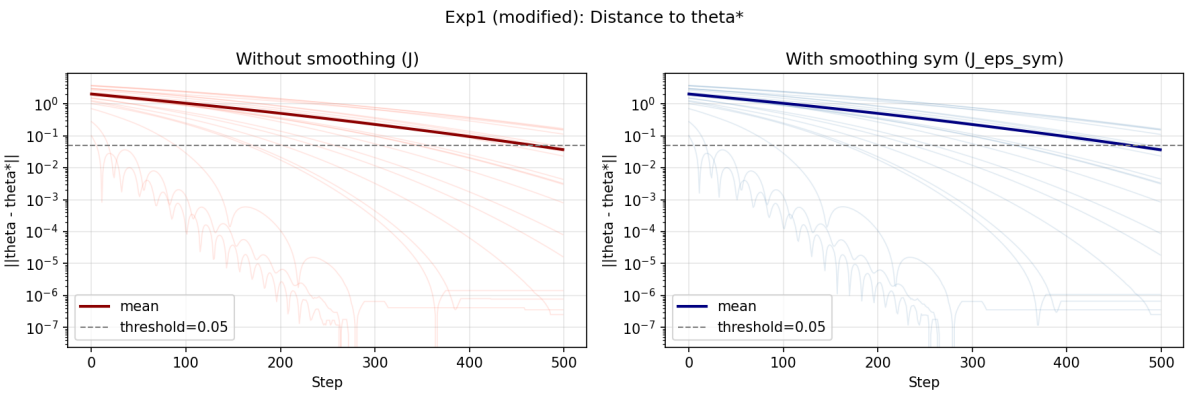

Fig. A2. *Distance ‖θ − θ*‖ over steps: raw J vs J_ε,sym.*

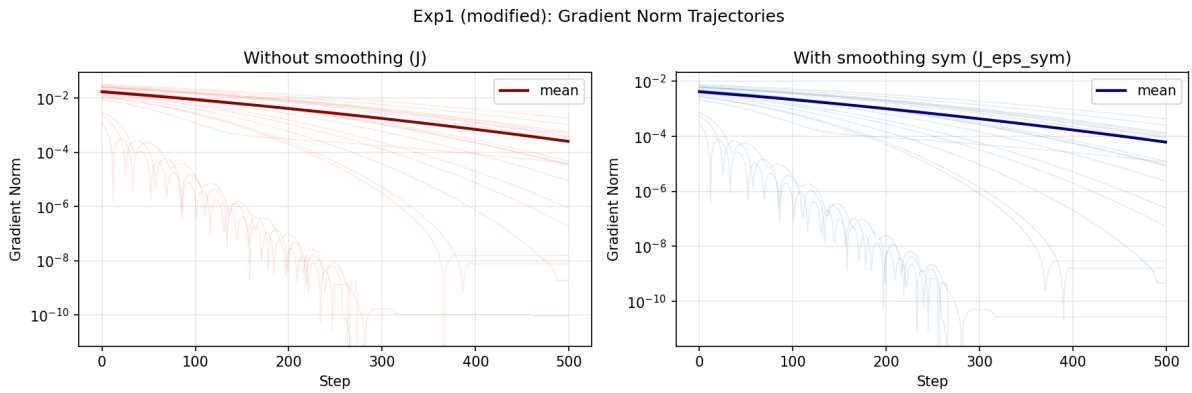

Fig. A3. *Gradient norm trajectories: raw J vs J_ε,sym.*

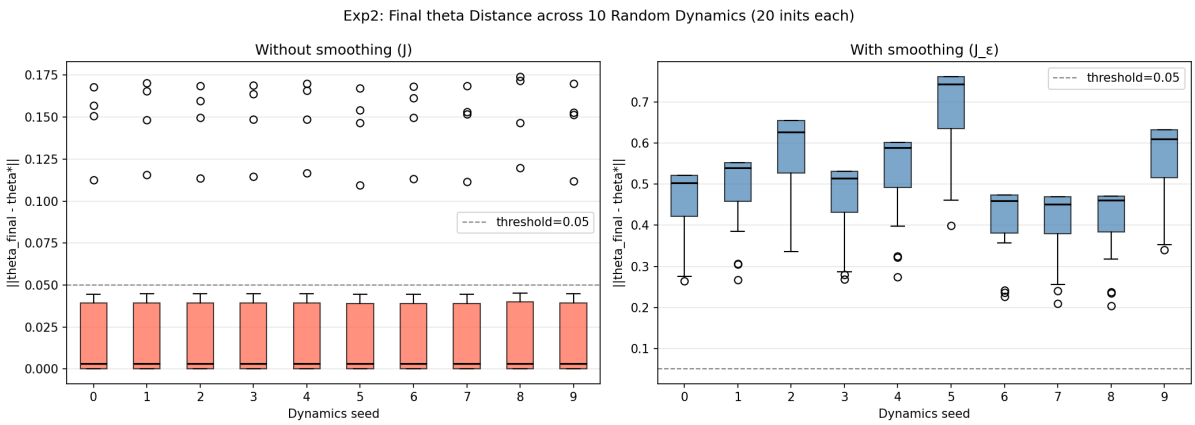

Fig. A5. *Final ‖θ_ε* − θ*‖ across 10 dynamics seeds (Exp2).*

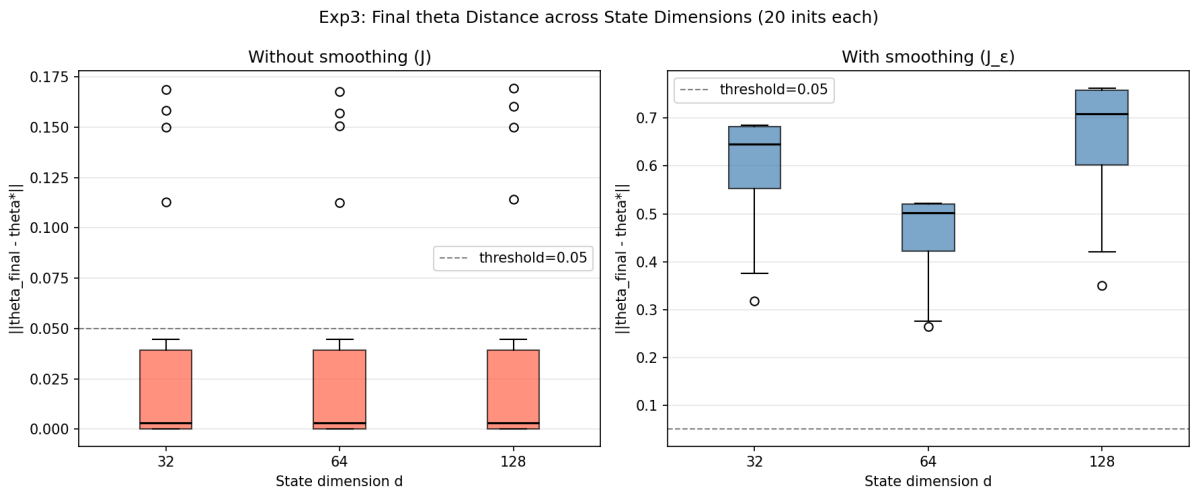

Fig. A7. *Final ‖θ_ε* − θ*‖ at d = 32/64/128 (Exp3).*

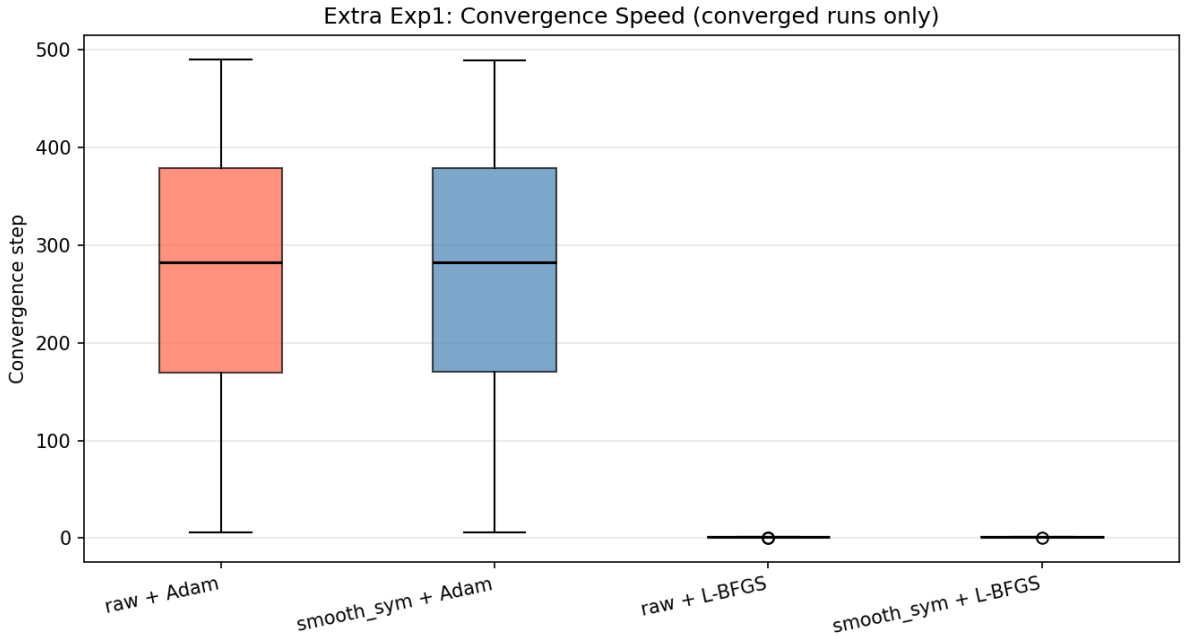

Fig. A8. *Convergence step distribution (boxplot): 4 conditions, Adam vs L-BFGS.*

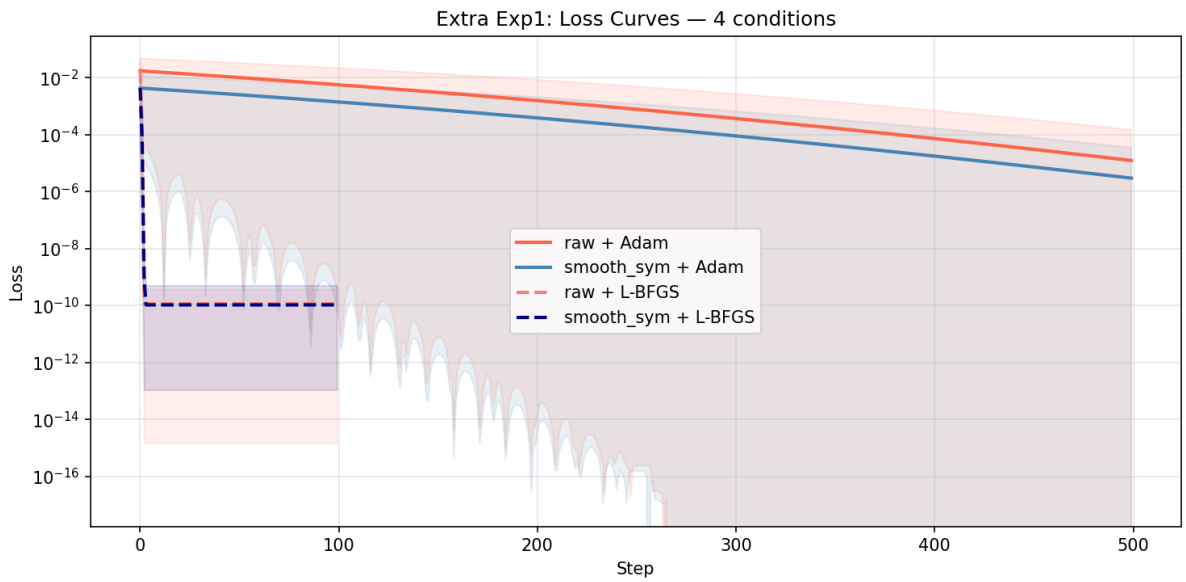

Fig. A9. *Loss curves: 4 conditions (raw/smooth_sym × Adam/L-BFGS).*

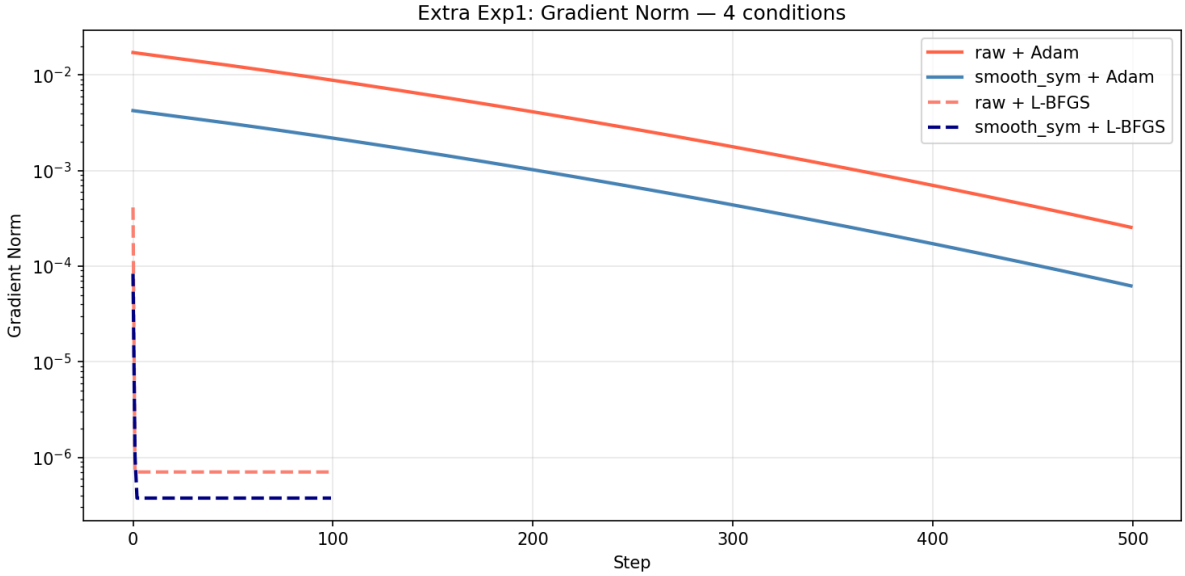

Fig. A10. *Gradient norm trajectories: 4 conditions.*